CSE315:Introduction to Data Science

WEEK-2

What is Data?

- The number of people who are not familiar with the term data is probably very low at present. In fact, the world is running on data, although data is not something that can be touched. Data is a small piece of information. Information is created by combining multiple data.
- “Data are symbols that represent properties of objects, events and their environments. They are products of observation” - Ackoff, Russell L

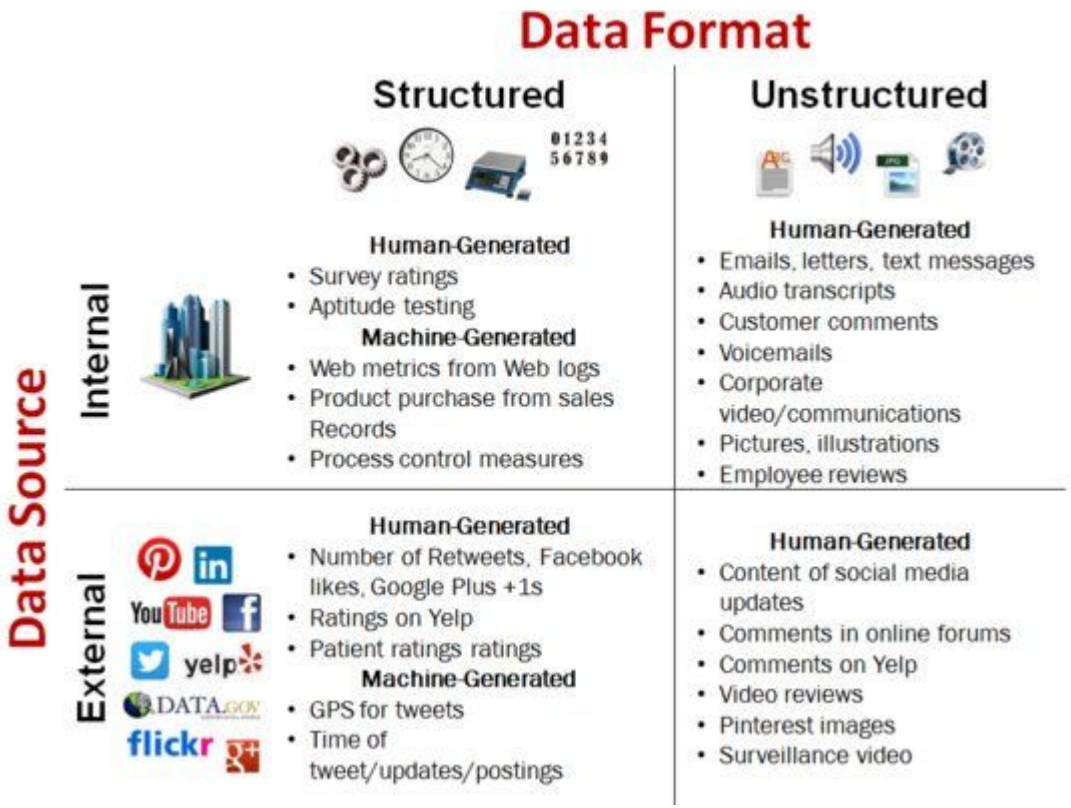
Information Extraction



Data Sources



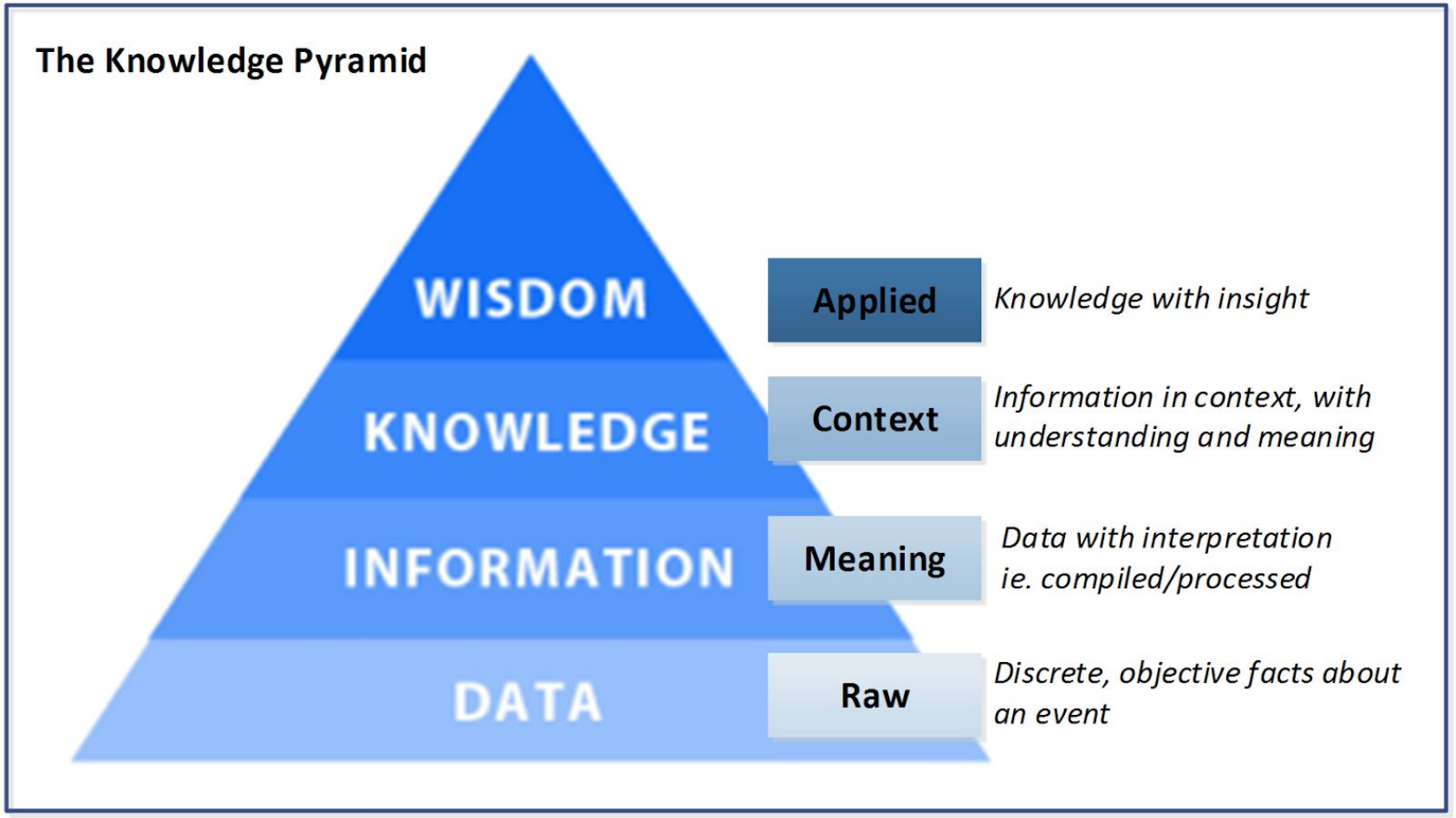
Data Format


Data Format
Structured
Unstructured
01234
56789
Data Source
Internal
Human-Generated
• Survey ratings
• Aptitude testing
Machine-Generated
• Web metrics from Web logs
• Product purchase from sales Records
• Process control measures
Human-Generated
• Emails, letters, text messages
• Audio transcripts
• Customer comments
• Voicemails
• Corporate video/communications
• Pictures, illustrations
• Employee reviews
External
YouTube
yelp
DATA.GOV
flickr
Human-Generated
• Number of Retweets, Facebook likes, Google Plus +1s
• Ratings on Yelp
• Patient ratings ratings
Machine-Generated
• GPS for tweets
• Time of tweet/updates/postings
Human-Generated
• Content of social media updates
• Comments in online forums
• Comments on Yelp
• Video reviews
• Pinterest images
• Surveillance video

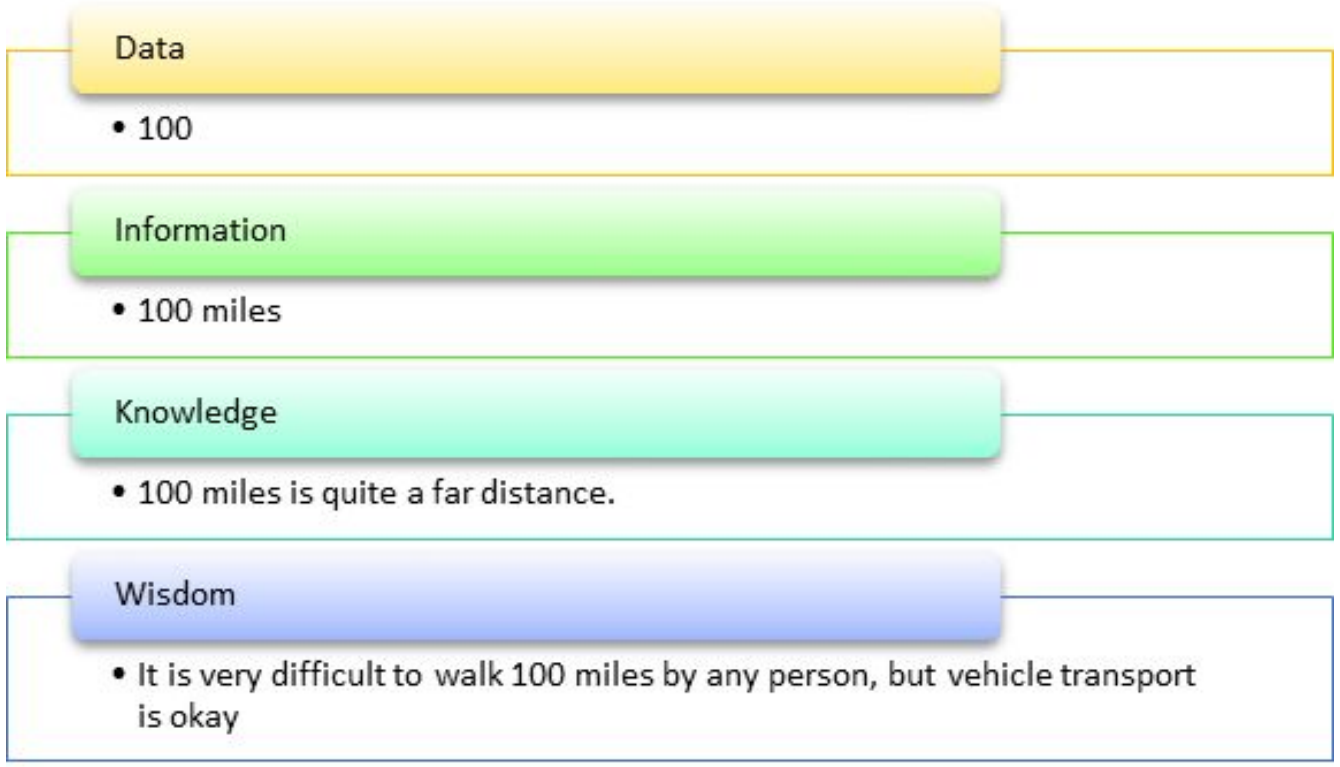
DIKW Pyramid

- DIKW is an acronym for Data, Information, Knowledge & Wisdom. This pyramid shows how small data can play a role in decision making.
- The DIKW pyramid is the continuum of the relationship from data to information, from knowledge to knowledge and from knowledge to wisdom.

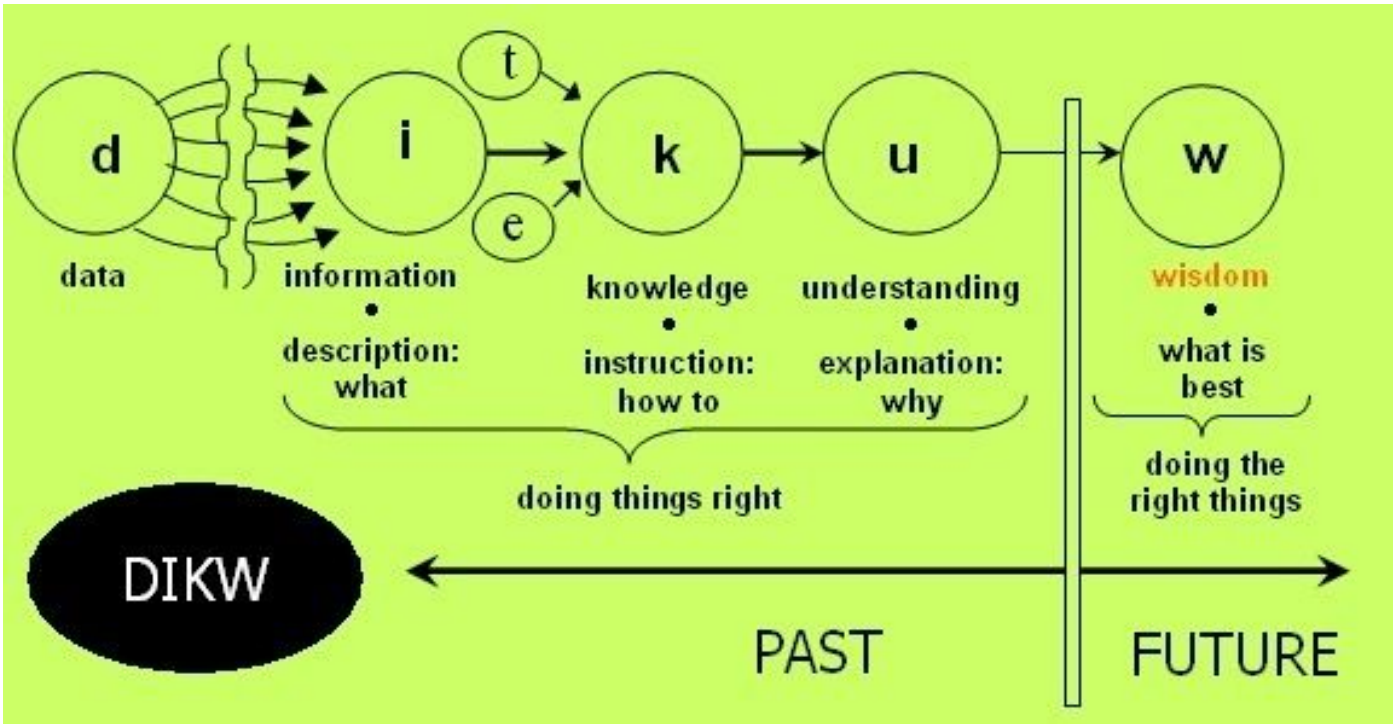
DIKW Pyramid



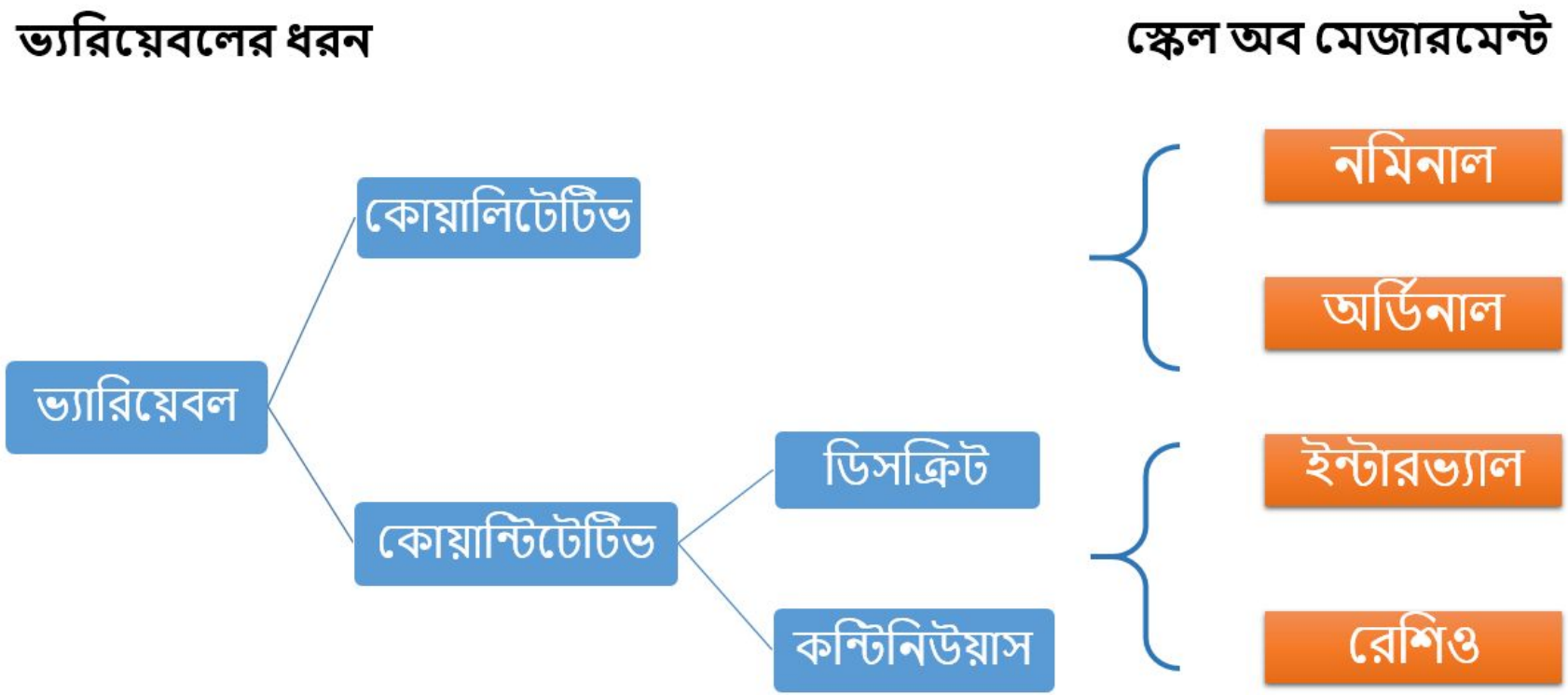
Data to Wishdom



Decision Making from Data



Types of Data



Quantitative Variable

- **Quantitative-** Quantitative Variable is basically a measurement or number based variable. Quantitative variables are also called numerical variables.
- Example: age (10 years, 26 years), weight (60 kg, 33 pounds), height (4 feet, 11 inches), distance (11 kilometers, 23 miles), etc. Such variables are expressed by numbers. 0

Types of Quantitative Variable

- Quantitative variables can be divided into two classes.
- **Discrete-** Discrete are integer numbers like 15 people, 5 cars etc. Such data can be counted or counted.
- For example, you can calculate how many people live in a house, and the number of people must be a whole number, that is, a house can have 4 people, 5 people, etc., but never 2.8 people, 3.5 people or Such a number does not live because the number of people is not possible in the form of fractions.

Types of Quantitative Variable

- **Continuous-** Continuous is the fractional size of the data.
- Example: human height 5 feet 4 inches is a fractional number. If you measure the height of the same person on a millimeter scale, you will get a few more millimeter fractions with 5 feet 4 inches.

Qualitative Variable

- **Qualitative** - Qualitative variable contains qualitative or characteristic data. Qualitative variables are also called categorical variables. Such as human hair color, gender, name, name of different places etc.

Scale of Measurement

- We can determine the scale of measurement based on the type of data that a variable contains, this scale of measurement is the nature of the data.

  - Nominal
  - Ordinal
  - Interval
  - Ratio

Nominal Data

- **Nominal** - Nominal scale can only be divided by category. Many people also call it simply 'label'.
- For example, the gender of a person, the color of something, the name of a person, object or place, etc. It is possible to count only on the nominal scale

Ordinal Data

- Ordinal - Ordinal scale contains orders. When it is possible to order any nominal data, it can be called ordinal.
- For example, a review of a restaurant's food (very good, good, fair, bad, very bad) is an ordinal scale, because if someone gives a 'very good' review, then on this scale it means the biggest while giving a 'very bad' review. Refers to small. If you think about the nominal scale, you can't order a person's hair color (white, black, gold), that's the difference between a nominal and an ordinal scale.

Interval Data

- **Interval** - The interval scale contains a certain distance or difference between data orders. This means that in addition to having all the features of the ordinal, this scale also has another feature called 'fixed distance'.
- **An excellent example** of this scale is the thermometer. Suppose the difference between the scales of a thermometer is 10 degrees Celsius then its clocks will be 0, 10, 20, 30, 40, 50, 60, 60, 60, 90, 100,110 etc. There is a certain distance or difference between these orders, the difference between each order is 10 degrees. This is the interval scale.

Ratio Data

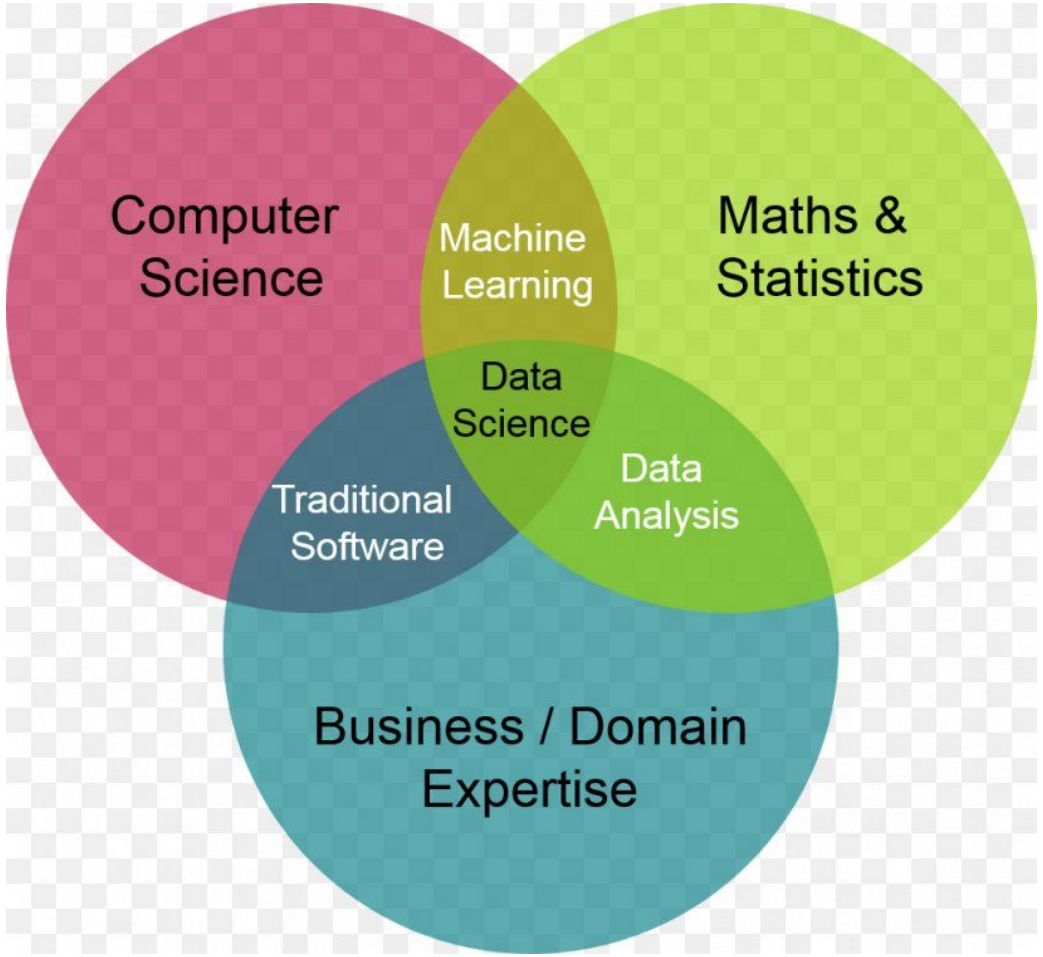
- **Ratio** - The biggest feature of the ratio scale is "Absolute Zero". Now the question is what is **Absolute Zero**?
- To answer this question we will use temperature measurement as an example. If you are asked, does 0 or zero degree temperature mean that there is no temperature? The answer is that even at a temperature of 0 or zero degrees the temperature remains the same because the measurement of temperature on the temperature scale can be minus. So even if the thermometer has a temperature of 0 degrees, you can't say there is no temperature! **So there is no absolute zero in temperature measurement.** Absolute zero is true emptiness, meaning zero is nothing.

Operations in Data

	Nominal	Ordinal	Interval	Ratio
Count	√	√	√	√
Mode	√	√	√	√
Order		√	√	√
Median		√	√	√
Mean			√	√
Add or Subtract			√	√
Multiplication or Division				√
Absolute Zero				√

Data Science Venn Diagram

Data science is the science of transformation of data into actionable insights



Different mysteries and Different patterns

Medical Dataset: Malignant or Benign

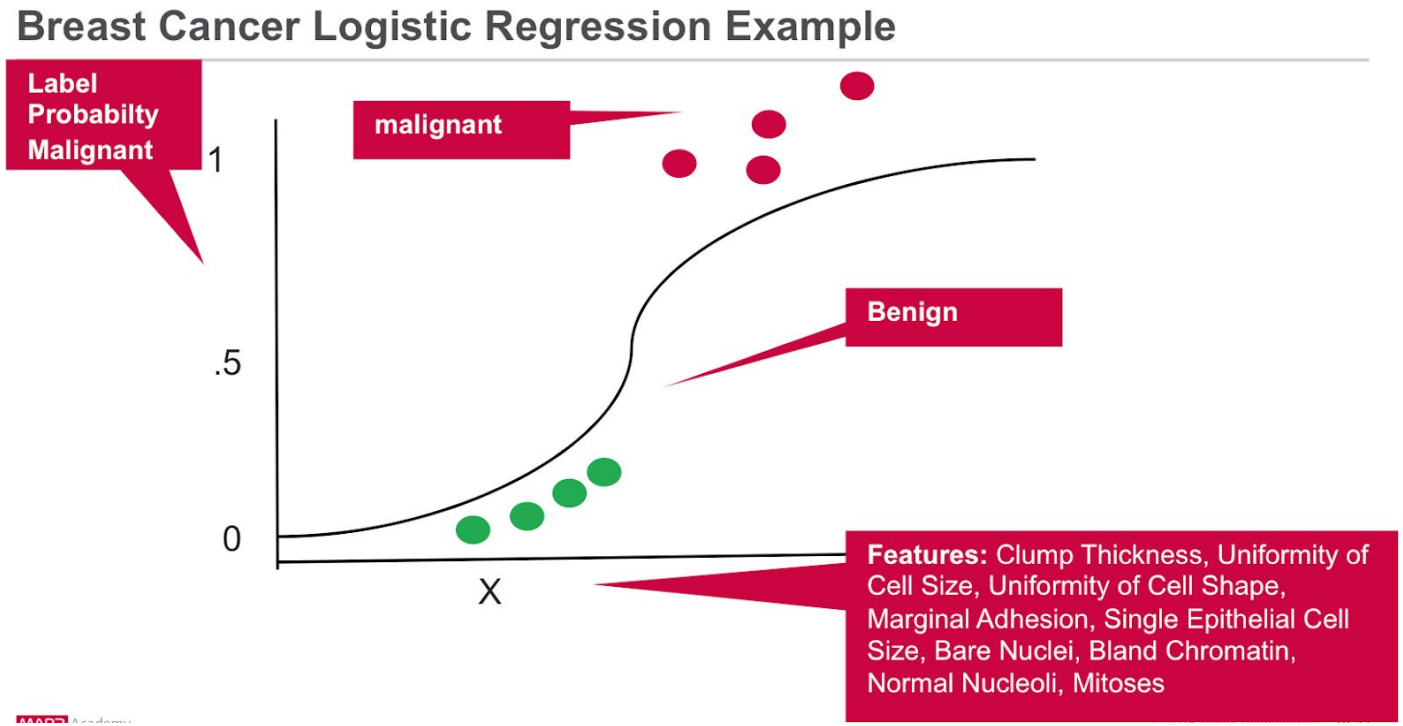
id	diagnosis	radius_me	texture_n	perimeter	area_mea	smoothne	compactn	concavity_	concave p	symmetry	fractal_dir	radius_se	texture_s	perimeter	area_se	smoothne	compactn	concavity_	concave p	symmetry	fractal_dir	radius_
842302	M	17.99	10.38	122.8	1001	0.1184	0.2776	0.3001	0.1471	0.2419	0.07871	1.095	0.9053	8.589	153.4	0.0064	0.04904	0.05373	0.01587	0.03003	0.00619	25
842517	M	20.57	17.77	132.9	1326	0.08474	0.07864	0.0869	0.07017	0.1812	0.05667	0.5435	0.7339	3.398	74.08	0.00523	0.01308	0.0186	0.0134	0.01389	0.00353	24
8.4E+07	M	19.69	21.25	130	1203	0.1096	0.1599	0.1974	0.1279	0.2069	0.05999	0.7456	0.7869	4.585	94.03	0.00615	0.04006	0.03832	0.02058	0.0225	0.00457	23
8.4E+07	M	11.42	20.38	77.58	386.1	0.1425	0.2839	0.2414	0.1052	0.2597	0.09744	0.4956	1.156	3.445	27.23	0.00911	0.07458	0.05661	0.01867	0.05963	0.00921	14
8.4E+07	M	20.29	14.34	135.1	1297	0.1003	0.1328	0.198	0.1043	0.1809	0.05883	0.7572	0.7813	5.438	94.44	0.01149	0.02461	0.05688	0.01885	0.01756	0.00512	22
843786	M	12.45	15.7	82.57	477.1	0.1278	0.17	0.1578	0.08089	0.2087	0.07613	0.3345	0.8902	2.217	27.19	0.00751	0.03345	0.03672	0.01137	0.02165	0.00508	15
844359	M	18.25	19.98	119.6	1040	0.09463	0.109	0.1127	0.074	0.1794	0.05742	0.4467	0.7732	3.18	53.91	0.00431	0.01382	0.02254	0.01039	0.01369	0.00218	22
8.4E+07	M	13.71	20.83	90.2	577.9	0.1189	0.1645	0.09366	0.05985	0.2196	0.07451	0.5835	1.377	3.856	50.96	0.00881	0.03029	0.02488	0.01448	0.01486	0.00541	17
844981	M	13	21.82	87.5	519.8	0.1273	0.1932	0.1859	0.09353	0.235	0.07389	0.3063	1.002	2.406	24.32	0.00573	0.03502	0.03553	0.01226	0.02143	0.00375	15
8.5E+07	M	12.46	24.04	83.97	475.9	0.1186	0.2396	0.2273	0.08543	0.203	0.08243	0.2976	1.599	2.039	23.94	0.00715	0.07217	0.07743	0.01432	0.01789	0.01008	15
845636	M	16.02	23.24	102.7	797.8	0.08206	0.06669	0.03299	0.03323	0.1528	0.05697	0.3795	1.187	2.466	40.51	0.00403	0.00927	0.01101	0.00759	0.0146	0.00304	19
8.5E+07	M	15.78	17.89	103.6	781	0.0971	0.1292	0.09954	0.06606	0.1842	0.06082	0.5058	0.9849	3.564	54.16	0.00577	0.04061	0.02791	0.01282	0.02008	0.00414	20
846226	M	19.17	24.8	132.4	1123	0.0974	0.2458	0.2065	0.1118	0.2397	0.078	0.9555	3.568	11.07	116.2	0.00314	0.08297	0.0889	0.0409	0.04484	0.01284	20
846381	M	15.85	23.95	103.7	782.7	0.08401	0.1002	0.09938	0.05364	0.1847	0.05338	0.4033	1.078	2.903	36.58	0.00977	0.03126	0.05051	0.01992	0.02981	0.003	16
8.5E+07	M	13.73	22.61	93.6	578.3	0.1131	0.2293	0.2128	0.08025	0.2069	0.07682	0.2121	1.169	2.061	19.21	0.00643	0.05936	0.05501	0.01628	0.01961	0.00809	15
8.5E+07	M	14.54	27.54	96.73	658.8	0.1139	0.1595	0.1639	0.07364	0.2303	0.07077	0.37	1.033	2.879	32.55	0.00561	0.0424	0.04741	0.0109	0.01857	0.00547	17
848406	M	14.68	20.13	94.74	684.5	0.09867	0.072	0.07395	0.05259	0.1586	0.05922	0.4727	1.24	3.195	45.4	0.00572	0.01162	0.01998	0.01109	0.0141	0.00209	19
8.5E+07	M	16.13	20.68	108.1	798.8	0.117	0.2022	0.1722	0.1028	0.2164	0.07356	0.5692	1.073	3.854	54.18	0.00703	0.02501	0.03188	0.01297	0.01689	0.00414	20
849014	M	19.81	22.15	130	1260	0.09831	0.1027	0.1479	0.09498	0.1582	0.05395	0.7582	1.017	5.865	112.4	0.00649	0.01893	0.03391	0.01521	0.01356	0.002	27
8510426	B	13.54	14.36	87.46	566.3	0.09779	0.08129	0.06664	0.04781	0.1885	0.05766	0.2699	0.7886	2.058	23.56	0.00846	0.0146	0.02387	0.01315	0.0198	0.0023	15
8510653	B	13.08	15.71	85.63	520	0.1075	0.127	0.04568	0.0311	0.1967	0.06811	0.1852	0.7477	1.383	14.67	0.0041	0.01898	0.01698	0.00649	0.01678	0.00243	1
8510824	B	9.504	12.44	60.34	273.9	0.1024	0.06492	0.02956	0.02076	0.1815	0.06905	0.2773	0.9768	1.909	15.7	0.00961	0.01432	0.01985	0.01421	0.02027	0.00297	10
8511133	M	15.34	14.26	102.5	704.4	0.1073	0.2135	0.2077	0.09756	0.2521	0.07032	0.4388	0.7096	3.384	44.91	0.00679	0.05328	0.06446	0.02252	0.03672	0.00439	18
851509	M	21.16	23.04	137.2	1404	0.09428	0.1022	0.1097	0.08632	0.1769	0.05278	0.6917	1.127	4.303	93.99	0.00473	0.01259	0.01715	0.01038	0.01083	0.00199	29
852552	M	16.65	21.38	110	904.6	0.1121	0.1457	0.1525	0.0917	0.1995	0.0633	0.8068	0.9017	5.455	102.6	0.00605	0.01882	0.02741	0.0113	0.01468	0.0028	26
852631	M	17.14	16.4	116	912.7	0.1186	0.2276	0.2229	0.1401	0.304	0.07413	1.046	0.976	7.276	111.4	0.00803	0.03799	0.03732	0.02397	0.02308	0.00744	22
852763	M	14.58	21.53	97.41	644.8	0.1054	0.1868	0.1425	0.08783	0.2252	0.06924	0.2545	0.9832	2.11	21.05	0.00445	0.03055	0.02681	0.01352	0.01454	0.00371	17

What do you understand from this data set?

What kind of data is inside? What is Insights? Is there a pattern?

The work of data science is to extract various information inside it.

Data Representation



Data Scientist's Profile

1. Computer science
2. Math
3. Statistics
4. Machine learning
5. Domain expertise
6. Communication and presentation skills
7. Data visualization

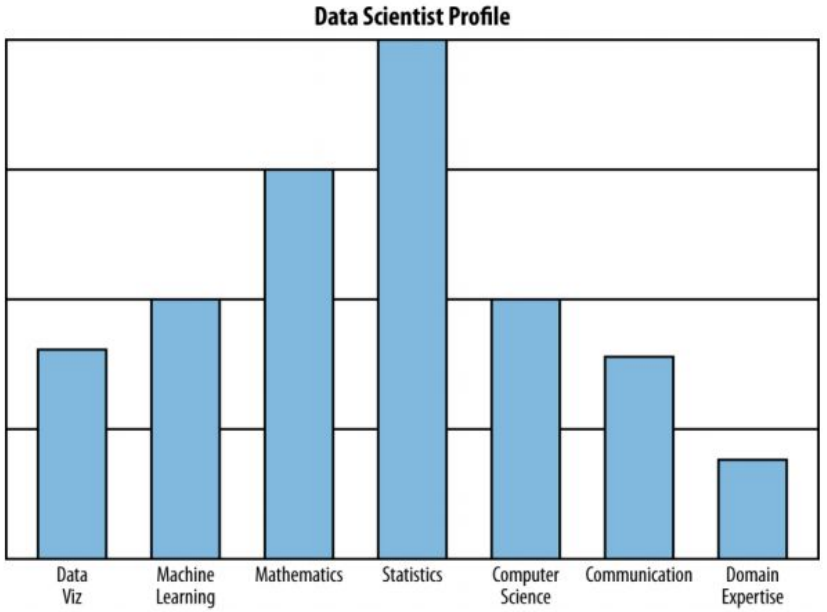
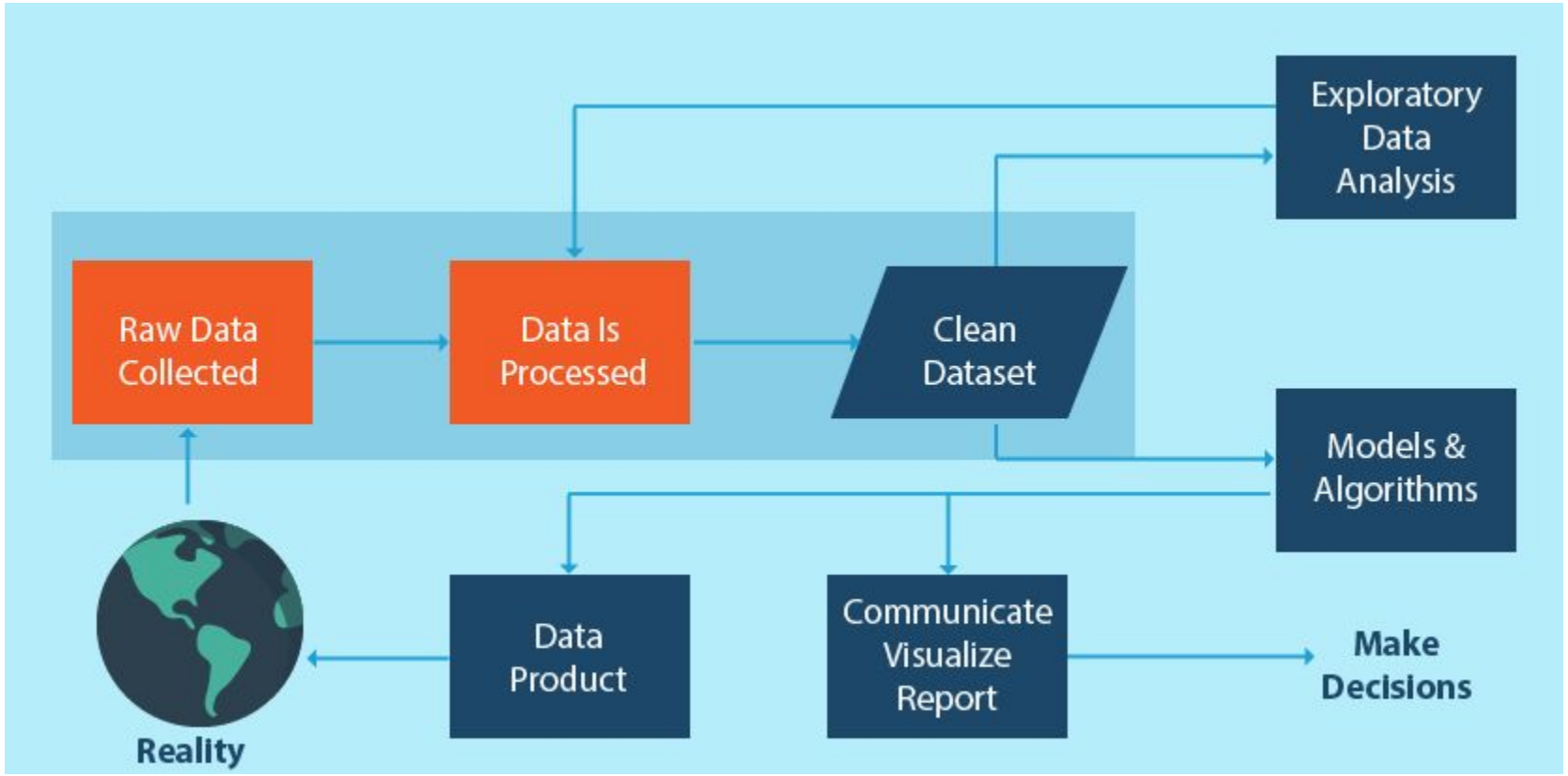


Figure 1: Rachel's Data Scientist Profile

Data Science Pipeline



Domain Knowledge

- The first step in the data science process is to know what we are going to do, that is, to determine what output or result we want to get from a data science project. In the case of data science project, the first step is to know what output the user wants through this project.

Domain Knowledge

- Another important factor in understanding the needs of the user is business understanding and domain knowledge. It is very important to know where the data we are going to work with comes from in the real world.
- That is, the process by which data is generated, how data measurement is done, ideas about all the attributes or features of data, etc. That is, the complete real-world approach to data and the idea of each feature is called Business Understanding and **Domain Knowledge**.

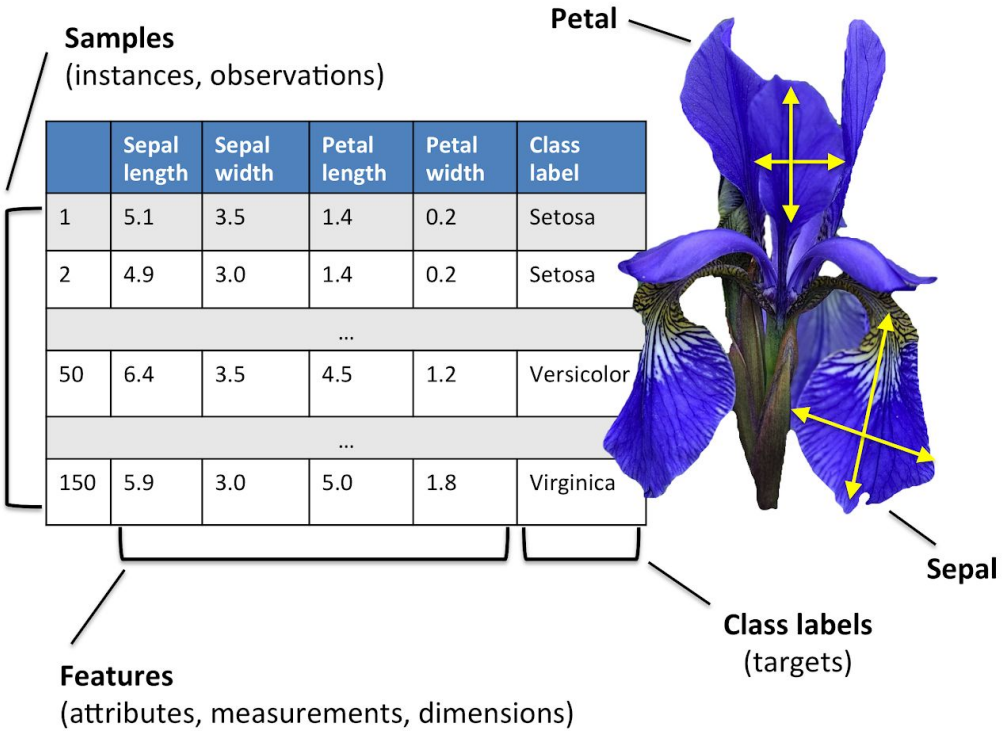
Example of IRIS Dataset

- We will clarify the concept of data science goals and domain knowledge through the world famous Iris data set. The father of the Iris data set is the statistician and biologist **R.A. Fisher**. Iris is the name of a flower. Iris Data Set is also called Iris Flower Data Set. Scientist **R.A. Fisher** published this data set in **1936**.



Iris Versicolor **Iris Setosa** **Iris Virginica**

Class Types and Attributes of IRIS Dataset



	Sepal length	Sepal width	Petal length	Petal width	Class label
1	5.1	3.5	1.4	0.2	Setosa
2	4.9	3.0	1.4	0.2	Setosa
			...		
50	6.4	3.5	4.5	1.2	Versicolor
			...		
150	5.9	3.0	5.0	1.8	Virginica